



Mo	Tu	We	Th	Fr	Sa	Su
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Terms

Date / /

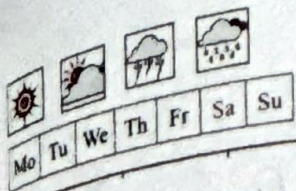
- ① SDV = Synthetic Data vault
- ② SMOTE = Synthetic minority oversampling technique.
- ③ CTGAN = Conditional Tabular Generative Adversarial Network

SMOTE ~~new~~ $x_{new} = x + \text{rand}(0,1) * (x_{\text{neighbour}} - x)$

$$x_{\text{new}} = x + \text{rand}(0,1) * (\text{neighbour} - x)$$

- ④ SMOTE-NC = ——— nominal & continuous
- ⑤ user prediction $\rightarrow$ o/p $\rightarrow \{0, 1, 2, 3, 4, 5\}$

user label = le. inverse transform(o/p)
(prediction)



SGP

Date / /

FLOW 1

Load Data



Encode



Baseline - SMOTE



Normalization



Baseline NN



TFL



Evaluate



web

FLOW 2

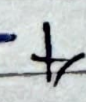
Load Data



Encode



~~Baseline~~ SMOTE-NC



Normalization



Baseline NN



TFL



Evaluate



web

young zenbow

OR

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Date / /

→ Borderline SMOTEL & SMOTE-NC

0 → Heat Dissipation

1 → No Failure

2 → Overstrain

3 → Power Failure

4 → Random Failure

5 → Tool wear

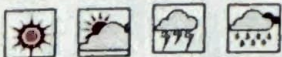
~~6 →~~

→ IMP

[y-label] column (drop when using data) OR (move to a variable).

[y-label] → in a variable X , will be used to inverse-transform the user's prediction.

Deep side



Mo Tu We Th Fr Sa Su

Date / /

* Raw 77% No Failure

* SMOTE BRD 33%

* SMOTE NC 16% — —

If MODEL does NOT work
as expected

Generate new Data

Because old data is generated,
with (Normalizing) before
SMOTE

Now apply SMOTE on {Maintenance - CSU}

then normalize Data.

→ No Generalization or Unseen
Data-



Mo	Tu	We	Th	Fr	Sa	Su
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Date / /

⇒ New Flow

data



Label encoding (categorical only)



SMOTE



Split (train/test)



Normalize



Modelling

⇒

Better

Modely

Date / /

⇒ Next

DL & Transfer learning.

⇒ ① ~~smote~~ ~~Booster~~

~~a) ML~~

~~random forest - job's~~

b) DL

Transformer

"[, none,]"

↓
batch

() feature - dim

seq - length → none = 1

for numerical.



Mo Tu We Th Fr Sa Su

1 Model

Date / /



SMOTE-Borderline

ML

* random-forest-99. joblib $\Rightarrow$ 99.97%

DL

* dl-baseline-hs (88.79, 27.53)

* dl-1-transformer-keras (88.9, 23.5)

* dl-2-transformer-keras
(92.97, 19.47)



Mo	Tu	We	Th	Fr	Sa	Su
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Date / /

SMOTE-NCML

* random-forest 98. joblib. => 98.06%

DL

* dl2-baseline. keras (89.29, 28.74)

* dl2-transformer. keras
(93.03, 20.98)



Mo	Tu	We	Th	Fr	Sa	Su
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Transformer

(Contd.)
Date _____

(Query, Key, Values).

→ How much each element should focus on others.

$$\rightarrow \text{Attention} \equiv \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

↳ dim. of key vector

⇒ Head → Perspective on data.